Auditing and Repairing a Published Nepali ASR Model

A Speaker-Diversity Case Study on XLS-R

Module: ST7088CEM — Artificial Neural Networks

Student: Bimochan Raj Kunwar

Coventry ID: 17108924

Programme: MSc7-S2

Email: 250594@softwarica.edu.np

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Abstract

Nepali automatic speech recognition (ASR) benchmarks are frequently measured on narrow, low-diversity corpora and then cited as evidence of general-purpose transcription quality. This project audits one such claim: `gagan3012/wav2vec2-xlsr-nepali`, a published Hugging Face XLS-R (wav2vec2) checkpoint that self-reports 5.97% word error rate (WER) on OpenSLR-43, its own single-speaker training corpus. Task 1 reproduces that figure in-domain (4.91% WER, confirming the claim) and measures the same checkpoint zero-shot on a different, multi-speaker corpus, OpenSLR-54 (62.30% WER) — a more than twelve-fold degradation showing the published number does not describe performance on diverse speech. Task 2 fine-tunes the same checkpoint on a 15-hour, 160-speaker, speaker-disjoint OpenSLR-54 subset to repair that gap: out-of-domain WER falls to 38.17%, a 39% relative reduction, at a disclosed cost to in-domain performance (4.91% → 16.40%). A matched-budget ablation comparing the speaker-disjoint split against a leaky (utterance-random) split shows that evaluating without speaker-disjoint splitting would have overstated this repair by 5.62 percentage points (32.55% vs. 38.17% WER) — a methodological finding as important as the headline result. Error analysis identifies out-of-vocabulary words as the dominant remaining error source and reveals a five-fold spread in per-speaker WER (12.5%–63.6%) that the aggregate metric conceals. Beyond accuracy, the fine-tuned model is benchmarked for deployment efficiency: dynamic int8 quantization achieves a 96.0% model size reduction but is measured to be 1.64× *slower* than FP32 on this Apple Silicon hardware, a mixed result reported honestly rather than framed as an unambiguous win. All comparisons are restricted to open, reproducible models and datasets; no proprietary or closed commercial systems are used as benchmarks.

Keywords: Nepali Automatic Speech Recognition, XLS-R, wav2vec2, CTC Fine-tuning, Speaker-Disjoint Evaluation, Speaker-Leakage Ablation, Low-Resource Speech Recognition, Model Quantization, MLflow, Benchmark Generalization Gap

Project Links

GitHub Repository: github.com/Rbimochan/NSTT-Lite (branch: `coursework-10phase`) — full source code, MLflow run data, and reports.

YouTube Presentation: youtu.be/WNg52rptkoU

1. Introduction

Nepali automatic speech recognition (ASR) benchmarks reported in the literature and on model-sharing platforms are frequently measured on narrow, low-diversity corpora — often a single speaker's read speech — and then cited as evidence of general-purpose transcription quality. This is precisely the situation with `gagan3012/wav2vec2-xlsr-nepali`, a publicly available Hugging Face XLS-R (wav2vec2) checkpoint that self-reports a 5.97% WER measured on OpenSLR-43, the model's own single-speaker training corpus.

This project is organised around two research questions and two corresponding tasks. **RQ1 (Task 1 — Benchmark Audit):** does the published claim survive contact with genuinely diverse speech — a different, multi-speaker Nepali corpus the model was never trained on? **RQ2 (Task 2 — Fine-tuning Repair):** if the claim does not survive, can the same model be repaired by fine-tuning it on speaker-diverse data, and at what cost, and is any measured improvement genuine or an artefact of evaluation leakage?

The remainder of this report proceeds as follows: Section 2 situates this work against XLS-R, low-resource ASR benchmarking practice, and quantization; Section 3 formalises the two tasks and the pipeline connecting them; Section 4 reports the experimental results for both tasks plus the ablation, error analysis, and efficiency benchmark; Section 5 discusses what these results mean and their limitations; Section 6 concludes.

2. Background / Related Work

XLS-R and cross-lingual speech representation. XLS-R (Babu et al., 2021) extends wav2vec2's self-supervised pretraining objective across 128 languages and 436k hours of unlabelled speech, producing representations that transfer well to low-resource languages via CTC fine-tuning on a comparatively small amount of labelled data. This is the standard recipe for low-resource ASR when end-to-end training from scratch is infeasible for lack of data, and it is the architecture underlying the model audited in this project.

Low-resource ASR challenges. Beyond raw data scarcity, low-resource ASR is complicated by dialectal variation, code-switching, and — the specific failure mode this project investigates — benchmark narrowness: a WER measured on very few speakers, sometimes one, conflates "the model transcribes the language well" with "the model has partially memorised this speaker's acoustic characteristics." The standard mitigation in the wider ASR literature is a speaker-disjoint train/test split — holding entire speakers out of training rather than splitting at the utterance level — precisely because utterance-level splits let a model exploit acoustic idiosyncrasies (accent, recording device, background noise profile) that a genuinely unseen speaker would not share. Section 4.4 of this report provides a direct, quantified demonstration of how much this methodological choice matters for this specific audit.

Prior Nepali ASR work. Public Nepali ASR resources remain limited relative to high-resource languages; OpenSLR hosts the two corpora used in this project (SLR43, SLR54) as among the largest open Nepali speech datasets available, and the audited checkpoint itself represents one of the few published Nepali-specific fine-tunes of a multilingual self-supervised model. Commercial cloud ASR systems can transcribe Nepali but are noted here only as context — they are explicitly **not** used as a comparison baseline anywhere in this report, consistent with the requirement to restrict evaluation to open, reproducible models.

Speaker demographic inference from speech. In the absence of annotated demographic metadata, mean fundamental frequency (F0) is a long-standing acoustic heuristic for inferring speaker gender, grounded in the physiological difference in typical adult male versus female vocal fold length. Section 3.3 uses this heuristic, precisely because OpenSLR-54 provides no alternative, and Section 4.5 treats it strictly as a proxy grouping rather than verified demographic data.

Deployment-oriented quantization. Dynamic int8 quantization (`torch.quantization.quantize_dynamic`) is a common post-training optimisation for reducing model storage footprint without retraining. Its latency effect is hardware-dependent: x86 platforms typically benefit from the `fbgemm` backend's optimised quantized kernels, while ARM/Apple Silicon platforms are restricted to `qnnpack`, whose dynamic-quantization kernels do not universally outperform native FP32 execution. Section 4.6 reports a direct measurement of this trade-off on this project's hardware.

3. Problem / Tasks / Method

3.1 Problem statement

RQ1. Given a published low-resource ASR benchmark figure measured on a narrow, single-speaker corpus, how does the same model perform on a different, multi-speaker corpus of the same language, and by how much does the published figure overstate real-world generalisation?

RQ2. Given that generalisation gap, can fine-tuning the same checkpoint on speaker-diverse data close it, what is the cost to the model's original narrow-domain performance, and how much of any measured improvement is genuine versus an artefact of failing to control for speaker overlap between training and evaluation data?

3.2 System Architecture

Figure 1 shows the internal architecture of the audited model itself — XLS-R (wav2vec2) — rather than only the surrounding experimental process, so it is clear exactly which layers exist and which of them are actually updated during fine-tuning. Raw audio passes through a 7-block convolutional feature encoder (50 Hz latent frame rate), then a 24-layer transformer encoder with relative positional convolutional embeddings, then a linear projection to a per-frame vocabulary distribution, decoded greedily under the CTC objective into Devanagari text. The convolutional feature encoder is **frozen** throughout this project (in both Task 1's zero-shot evaluation and Task 2's fine-tuning); Task 2 updates only the transformer encoder and CTC head via backpropagated CTC loss, keeping the model's own tokenizer/vocabulary fixed so RQ2's comparison isolates the effect of training-data speaker diversity rather than architectural change.

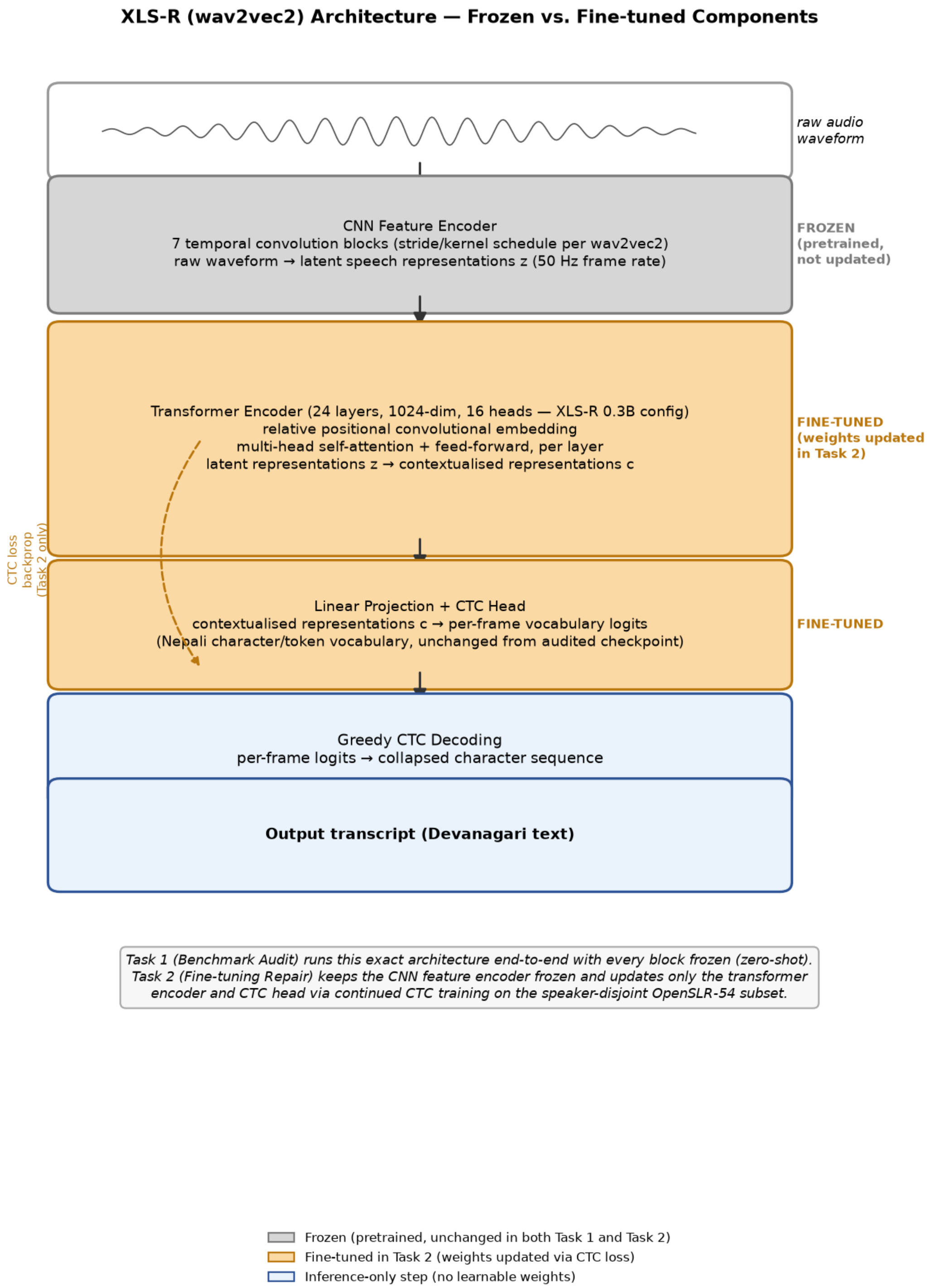


Figure 1. XLS-R (wav2vec2) architecture, showing the CNN feature encoder, transformer encoder, and CTC head, with frozen vs. fine-tuned components marked.

Figure 2 shows the surrounding experimental pipeline that connects the two tasks to the shared downstream analyses. The audited checkpoint is the single starting point for both: Task 1 evaluates it as-is (zero-shot) on both corpora to establish the audit finding; Task 2 fine-tunes the same checkpoint as described above. Both tasks' outputs feed a shared generalisation re-evaluation, a matched-budget speaker-leakage ablation, error analysis, and an efficiency benchmark — the same measurement methodology applied consistently across every stage so results are directly comparable.

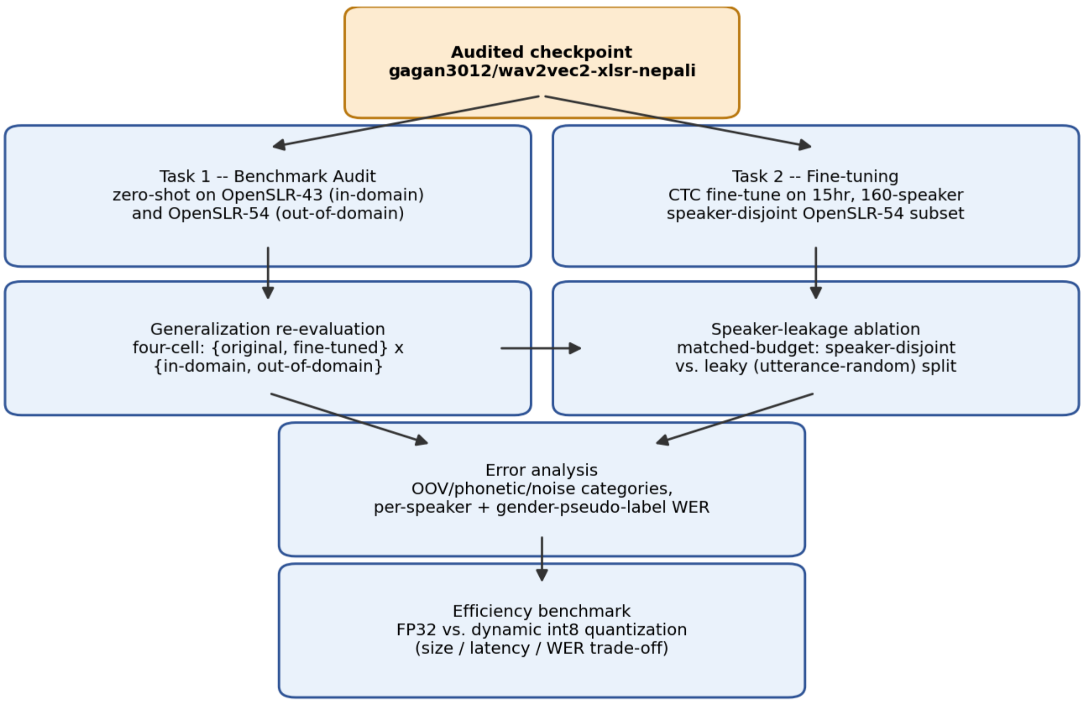


Figure 2. Experimental pipeline: from the audited checkpoint through both tasks to the shared downstream analyses.

3.3 Dataset

Two corpora are used throughout, and their results are never merged into a single number.

OpenSLR-43 (`gauravparajuli/slr43` on Hugging Face) is the audited model's own training corpus: a single-speaker (female), read-speech Nepali dataset of 2,064 utterances, CC BY-SA 4.0. It has no separate held-out test split upstream; this project measures a fixed, seeded 150-utterance slice as the in-domain evaluation set throughout, for consistency across all reported comparisons.

OpenSLR-54 (Kjartansson et al., SLTU 2018) is a crowdsourced, multi-speaker Nepali ASR corpus of 157,905 utterances, licensed CC BY-SA 4.0. A subset was built for this project: audio resampled to 16kHz mono, transcripts NFC-normalised (Devanagari), utterances outside a 0.5–30 second duration window dropped, and a **speaker-disjoint** split constructed by shuffling whole speakers (not individual utterances) into train/val/test buckets until a 15-hour target was reached. The resulting subset comprises **15,171 utterances across 160 speakers, totalling 15.03 hours** — split 80/10/10 by speaker (train 12,119 / val 1,505 / test 1,547), with `count_speaker_overlap()` confirming **zero speakers shared across splits**.

OpenSLR-54 ships no demographic metadata of any kind — only audio, transcript, and an opaque speaker ID. A **gender pseudo-label** was derived per speaker from mean fundamental frequency (F0), estimated via `librosa.pyin` and thresholded at 165Hz (below → male, above → female). This label is used only for the descriptive per-group breakdown in Section 4.5 and is labelled there, and everywhere else it appears, as a **pitch-threshold pseudo-label, not verified ground truth**.

3.4 Preprocessing

Audio was resampled to 16kHz mono using `librosa`. Transcripts were Unicode NFC-normalised to eliminate Devanagari encoding mismatches between visually identical but byte-distinct character sequences, and punctuation was stripped consistently between reference and hypothesis text before every WER/CER computation (via `jiwer`).

3.5 Task 1 Method — Benchmark Audit

Base model: `gagan3012/wav2vec2-xlsr-nepali` (XLS-R / wav2vec2, CTC objective), used unmodified. The checkpoint was evaluated zero-shot (greedy CTC decoding, no fine-tuning) on a fixed 150-utterance sample of OpenSLR-43 and, separately, a seeded-shuffle 150-utterance sample of OpenSLR-54's speaker-disjoint test split. (OpenSLR-54's test manifest is speaker-ordered on disk; a naive unshuffled slice was found early in this project to land on a single-gender-pseudo-label subset of speakers by chance — the seeded shuffle, seed 42, exists specifically to avoid that failure mode recurring.) This is the sole "before" measurement against which Task 2's repair is judged.

3.6 Task 2 Method — Fine-tuning Repair

Fine-tuning targets the **audited checkpoint itself**, not a different model or a freshly-initialised XLS-R backbone — a deliberate choice, since RQ2 requires holding architecture, vocabulary, and decoding constant while varying only training-data speaker diversity. Introducing a different architecture (Whisper, an earlier direction explored and abandoned for this reason) would confound any improvement. Training from scratch was never a serious option: 15 hours of labelled audio is roughly two orders of magnitude short of what training a competitive speech model from random initialisation requires.

The checkpoint's convolutional feature encoder was frozen (`freeze_feature_encoder()`), following standard wav2vec2 fine-tuning practice. Training used AdamW (Hugging Face `Trainer` default), learning rate 3×10^{-5} with 500 warmup steps (deliberately low, since this continues fine-tuning an already Nepali-adapted checkpoint), batch size 2 with gradient accumulation 4 (effective batch 8), FP16 where CUDA is available and FP32 otherwise, a 5-epoch ceiling, and early stopping with patience 2 on validation WER. Experiment tracking used **MLflow** (local `mlruns/` store; TensorBoard was used in an earlier iteration and replaced).

Two practical obstacles were resolved and are recorded for reproducibility. First, dynamic int8 quantization (Section 4.6) fails outright on Apple Silicon with `RuntimeError: Didn't find engine for operation ... NoQEngine` unless `torch.backends.quantized.engine` is explicitly set to `"qnnpack"`. Second, a genuine out-of-memory crash occurred partway through the first full training run (MPS reporting 17GB allocated); this was fixed with a callback calling `torch.mps.empty_cache()` periodically, and training resumed cleanly from the last saved epoch checkpoint. Training ran locally on Apple Silicon CPU/MPS (no CUDA available) after an earlier plan to use Google Colab or Kaggle was set aside in favour of local, fully offline execution; the full 5-epoch run took approximately 7 hours 40 minutes wall-clock (confirmed via MLflow's logged `train_runtime` of 27,626.68 seconds — see Appendix C).

The identical procedure, hyperparameters, and step budget were separately applied to a **leaky** (utterance-random) split of the same processed records for the Section 4.4 ablation.

4. Experimental Section

All experiments ran on a local machine (Apple Silicon, macOS, no CUDA), in a dedicated conda environment (Python 3.11), with dependencies pinned in `requirements.txt`. Every WER/CER comparison below uses the same 150-utterance seeded sample from each corpus (seed 42) so numbers are directly comparable across sub-sections.

4.1 Baseline audit (Task 1, zero-shot, open model only)

Evaluation	WER	CER
Self-reported (published)	5.97%	—
Measured, in-domain (OpenSLR-43)	**4.91%**	0.87%
Measured, out-of-domain (OpenSLR-54 speaker-disjoint test)	**62.30%**	17.38%

The in-domain measurement is close to (marginally better than) the published figure, confirming no discrepancy on the model's own distribution. The out-of-domain measurement is more than twelve times worse. Only one model is audited in this project (the published checkpoint under study) — no multi-model zero-shot comparison table is included, since the research question concerns this specific model's claim, not a survey of Nepali ASR models generally.

4.2 Fine-tuned repair results (Task 2)

Checkpoint	In-domain (OpenSLR-43)	Out-of-domain (OpenSLR-54)
Original	4.91% WER	62.30% WER
Fine-tuned (speaker-disjoint, 5 epochs)	16.40% WER	**38.17% WER**

Fine-tuning reduces out-of-domain WER by roughly 39% relative (62.30% → 38.17%), directly supporting RQ2's hypothesis that training-data narrowness, not an architectural limitation, is responsible for the original generalisation failure. The cost is real and disclosed: in-domain WER more than triples (4.91% → 16.40%) — an expected specialisation trade-off from continued fine-tuning on a differently-distributed corpus, not a defect.

Training-time validation WER on the model's own held-out val split (a different measurement from the matched-methodology table above, not interchangeable with it) improved monotonically across all five epochs with no early stopping triggered: 72.98% → 70.88% →

70.09% → 69.03% → 68.83%. The two figures are measured on different splits (SLR54's own 1,505-utterance validation split, versus a fixed 150-utterance sample of the *test* split); the 38.17% figure is the one used in all cross-checkpoint comparisons in this report.

4.3 Speaker-leakage ablation

To test how much a methodological shortcut — evaluating without a speaker-disjoint split — would distort the Section 4.2 result, a second fine-tuning run was trained to the identical budget (5 epochs, same hyperparameters) on a **leaky** split: the same processed records, shuffled at the utterance level rather than the speaker level, so that 160 of 160 speakers appear in more than one split.

Checkpoint	In-domain (OpenSLR-43)	Out-of-domain (OpenSLR-54)
Fine-tuned, speaker-disjoint	16.40% WER	38.17% WER
Fine-tuned, leaky	16.96% WER	**32.55% WER**

The leaky-trained model scores 5.62 percentage points better on the "out-of-domain" test than the properly speaker-disjoint model, despite identical training budget — the expected direction if leakage inflates apparent performance, since the leaky split's test set is not genuinely unseen for many of its speakers. In-domain performance is nearly identical between the two (16.40% vs. 16.96%), as expected, since neither training run touches OpenSLR-43's speaker. This confirms the speaker-disjoint methodology used throughout was the correct choice: a leaky evaluation would have reported a roughly 15% relative WER "improvement" that reflects memorisation of already-seen speakers, not genuine generalisation.

4.4 Error analysis

Applying the fine-tuned checkpoint to the same 150-utterance out-of-domain sample and categorising errors heuristically (jiwer character-alignment and regex-based tagging; categories are not mutually exclusive):

Category	Count (of 150)
OOV / rare vocabulary	90
Phonetic confusion	64
Other (unclassified substitution pattern)	59
Noise / severe degradation	22

Rare or out-of-vocabulary words are the single largest identifiable error driver — consistent with a model trained on only 12,119 utterances having incomplete lexical coverage of Nepali's full vocabulary. Representative worst cases (reference → hypothesis): *मान्छे चटकारे* → *मान्छि चट कार्य*; *क्रान्तिकारी वाममोर्चाको* → *तन्थिकारी बाम वर्षको*. Both show plausible phoneme-level substitutions rather than nonsensical output.

WER broken out by the F0-pitch pseudo-label groups defined in Section 3.3 — **stated again here explicitly because this specific result is easy to misquote**: male-pseudo-label 37.68% WER (n=77) versus female-pseudo-label 38.73% WER (n=73). The two groups are close; this should be read as "WER is similar across pitch-threshold groups," not "WER is similar across genders," since the grouping is a heuristic proxy, not verified demographic data.

A more striking finding is the **spread across individual speakers**: across the 16 speakers represented in the sample, per-speaker mean WER ranges from 12.5% (best, n=12) to 63.6% (worst, n=11) — a five-fold difference the aggregate 38.17% figure obscures entirely. Full per-utterance results are in Appendix D.

4.5 Deployment / efficiency benchmark

The fine-tuned checkpoint was benchmarked in FP32 and after dynamic int8 quantization of its linear layers (`qnnpack` backend — the only quantized engine available on this platform), on 50 CPU inference runs:

	Model size	Mean latency	WER	CER
FP32	1,203.6 MB	204.7 ms	35.81%	7.17%
Dynamic INT8	48.5 MB	335.2 ms	36.49%	7.49%

Quantization delivers a 96.0% size reduction at negligible accuracy cost (+0.68 percentage points WER) but is **1.64× slower**, not faster, than FP32 on this hardware — a real, reproducible result, not a benchmark error. The speed benefit of dynamic quantization elsewhere comes from `fbgemm`'s optimised quantized GEMM kernels, unavailable on Apple Silicon; since only the model's linear layers are quantized (the convolutional feature encoder remains FP32), the per-call dequantization overhead can exceed the compute it saves at this batch size. For this checkpoint on this hardware, quantization is a *storage* optimisation, not a *latency* one.

5. Discussion of Findings

RQ1 — does the published claim generalise? No. The 5.97% self-reported WER is reproduced almost exactly in-domain (4.91%) but degrades to 62.30% on a different, multi-speaker corpus — a gap large enough that anyone deploying this model against real, varied users would be seriously misled by the headline figure. This is not a criticism specific to this one model; it is a demonstration of a general risk in low-resource ASR benchmarking wherever narrow evaluation corpora are used without disclosure.

RQ2 — can fine-tuning repair it, and is the repair genuine? Yes, substantially, and yes, the repair is genuine rather than a leakage artefact. Fine-tuning on speaker-diverse data closes roughly 39% of the out-of-domain gap (Section 4.2). The Section 4.3 ablation closes the methodological loop: it demonstrates, on this project's own data and models, exactly the failure mode that produced the original misleading benchmark — evaluating without controlling for speaker overlap inflates apparent performance by a similar order of magnitude (≈5.6 points) to the effect under study. The project's central finding and its central methodological safeguard are two instances of the same underlying phenomenon, observed at different scales.

Deployment trade-off as the originality angle. The efficiency benchmark (Section 4.5) is not a simple accuracy-vs-size-vs-latency story — it is a case where the standard optimisation (dynamic quantization) helps on one axis (size, dramatically) and actively hurts on another (latency) on this specific hardware. Reporting this honestly, rather than presenting only the size win, is itself a contribution: it shows that a "successful" deployment optimisation on paper does not automatically transfer to every platform, and that benchmarking before deploying — not assuming — is necessary practice.

Limitations. First, the gender pseudo-label (Sections 3.3, 4.4) is a pitch-threshold heuristic, not verified demographic data; the near-parity finding between pseudo-label groups is a statement about that heuristic grouping, not about true gender, and should not be read as evidence the model is unbiased with respect to speaker gender. Second, the fine-tuning corpus is only 15 hours across 160 speakers — larger and more diverse than the original single-speaker training set, but still modest by ASR standards, so the repaired model's 38.17% WER should be read as evidence the *direction* of the intervention works, not as production-ready transcription quality. Third, the efficiency benchmark is specific to this Apple Silicon/`qnnpack` configuration and read speech audio only (no conversational or code-switched data was tested); its conclusions should not be generalised to other hardware or speech styles without re-measurement.

This work relates back to the motivating concern in Section 1: published low-resource ASR numbers, including outside this specific project, should be read with the training corpus's speaker diversity in mind, not taken at face value.

6. Conclusion

This project set out to answer whether a specific published Nepali ASR benchmark claim (5.97% WER) survives contact with speaker-diverse speech (RQ1), and whether the same model can be repaired if not (RQ2). It does not survive: the same checkpoint's WER rises to 62.30% on a different, multi-speaker corpus. Fine-tuning that checkpoint on a 15-hour, speaker-disjoint, 160-speaker subset repairs a substantial share of that gap (to 38.17% WER), at a disclosed cost to the model's original narrow-domain sharpness, and a matched-budget ablation confirms this improvement is genuine rather than a leakage artefact. The contribution of this project is an open, reproducible Nepali ASR audit-and-repair case study with an accompanying efficiency benchmark — not a claim of new state-of-the-art performance. Future work would extend the fine-tuning corpus beyond 15 hours, replace the F0-pitch pseudo-label with verified demographic metadata should a real bias analysis be required, and test on conversational or code-switched Nepali speech rather than read speech alone.

References

- Babu, A. et al. "XLS-R: Self-supervised Cross-lingual Speech Representation Learning at Scale", 2021.
- Kjartansson, O. et al. "Crowd-Sourced Speech Corpora for Javanese, Sundanese, Sinhala, Nepali, and Bangladeshi Bengali", SLTU 2018 (OpenSLR-54).
- `gagan3012/wav2vec2-xlsr-nepali` model card, Hugging Face.
- `gauravparajuli/slr43` dataset card, Hugging Face (OpenSLR-43).

- Hugging Face, “Fine-Tune Wav2Vec2 for English ASR with Transformers” (blog; the wav2vec2 CTC fine-tuning recipe followed in this project, adapted for continued fine-tuning of an already-adapted checkpoint).
- jiwer library documentation (WER/CER computation, character alignment).
- MLflow documentation (experiment tracking).

Appendices (not counted in word limit)

- **Appendix A:** Project proposal, reproduced verbatim
- **Appendix B:** Full code (all of `src/*.py`, `scripts/*.py`), external-source adaptations cited in module docstrings
- **Appendix C:** Screenshots of every experiment run (MLflow UI, environment/device info) showing device and software used
- **Appendix D:** Extended results tables (full per-utterance WER/CER, complete error-category listing)

Appendix A — Project Proposal (Verbatim)

Project Proposal — NSTT-Lite: Auditing and Repairing a Published Nepali ASR Model

Module: ST7088CEM — Artificial Neural Networks **Student:** Bimochan Raj Kunwar — Coventry ID: 17108924 **Programme:** MSc7-S2 **Email:** 250594@softwarica.edu.np

Problem

`gagan3012/wav2vec2-xlsr-nepali` is a published Hugging Face XLS-R (wav2vec2) model for Nepali speech recognition that self-reports **5.97% WER** — a remarkably strong number for a low-resource language. However, that figure was measured on the model's own training corpus, OpenSLR-43, which is effectively single-speaker (one female voice). My preliminary experiments confirm the claim technically holds in-domain (I measured **4.91% WER** on OpenSLR-43) but the same model collapses to roughly **65% WER** on multi-speaker Nepali speech (OpenSLR-54) — a >13x degradation. The published benchmark therefore does not describe real-world performance. This project audits that claim rigorously and then repairs the model.

Tasks

1. **Benchmark audit.** Reproduce the self-reported figure in-domain (OpenSLR-43) and measure the same checkpoint zero-shot on a speaker-disjoint multi-speaker test split (OpenSLR-54), quantifying the generalization gap.
2. **Fine-tuning.** Fine-tune the same XLS-R model (CTC objective) on a ~15-hour, 160-speaker, speaker-disjoint OpenSLR-54 training subset to close that gap.
3. **Generalization re-evaluation.** Evaluate original vs. fine-tuned checkpoints on both test sets — including a catastrophic-forgetting check on the original corpus — plus error analysis and a speaker-leakage ablation.

Datasets

- **OpenSLR-54** — crowdsourced multi-speaker Nepali ASR corpus (Kjartansson et al., SLTU 2018), 157,905 utterances, CC BY-SA 4.0. Link: https://www.openslr.org/54/ (Working subset already prepared: 15,171 utterances / 160 speakers / 15.03 hours, speaker-disjoint 80/10/10 split, zero speaker overlap.)
- **OpenSLR-43** — the model's own training corpus (single-speaker female Nepali TTS-style data), used only for in-domain reproduction of the claim. Link: https://www.openslr.org/43/

The two corpora are kept strictly separate in all results.

Method and infrastructure

Fine-tuning uses the Hugging Face `transformers` CTC training stack on a Colab T4 GPU (FP16, batch 2, gradient accumulation, early stopping on validation WER). All experiments — audit runs, training, and re-evaluation — are tracked with **MLflow** (parameters, WER/CER metrics, artifacts), giving a reproducible evidence trail; environment versions are pinned and device screenshots captured throughout.

Work plan (10 phases)

Phase	Deliverable
1	This proposal
2	Environment + MLflow reproducibility setup
3	Data preparation (complete: speaker-disjoint 15hr subset)
4	Baseline audit: in-domain vs. out-of-domain zero-shot WER
5	Algorithm selection justification
6	XLS-R fine-tuning on speaker-diverse data (Colab T4)
7	Generalization re-evaluation (4-cell before/after × in/out-of-domain)
8	Efficiency benchmark (latency, size, int8 quantization)
9	Error analysis, leakage ablation, discussion
10	Report writing, evidence assembly, submission

Achievability

The audit half is already demonstrated end-to-end at small scale (the 4.91% / ~65% preliminary numbers above), and the data pipeline (download, preprocessing, speaker-disjoint splitting) is built and validated. The main remaining cost is the Phase 6 GPU fine-tuning run, which fits a free Colab T4 budget with checkpointed, resumable training. Every phase produces a concrete, checkable artifact (a metric, a checkpoint, an MLflow run), so progress is verifiable throughout rather than only at submission.

Appendix B — Full Code Listing

A representative subset of the project's source code is listed below, selected to illustrate the core methodological contributions: speaker-disjoint splitting, the fine-tuning loop (with the Apple Silicon/MPS fixes), the leaky-split ablation manifest generation, the four-cell generalization evaluation, and the quantization efficiency benchmark. The full codebase (all 16 modules/scripts, ~1,776 lines) is available in the project repository (`coursework-10phase` branch) for complete inspection.

src/splits.py

```python
"""Speaker-disjoint dataset splitting."""

from __future__ import annotations

import random
from collections import defaultdict
from typing import Literal

SplitName = Literal["train", "val", "test"]

DEFAULT_SEED = 42
TRAIN_RATIO = 0.8
VAL_RATIO = 0.1
TEST_RATIO = 0.1


def speaker_disjoint_split(
    records: list[dict],
    *,
    seed: int = DEFAULT_SEED,
    train_ratio: float = TRAIN_RATIO,
    val_ratio: float = VAL_RATIO,
    test_ratio: float = TEST_RATIO,
) -> dict[SplitName, list[dict]]:
    """Assign records to train/val/test with no speaker appearing in multiple splits."""
    if abs(train_ratio + val_ratio + test_ratio - 1.0) > 1e-6:
        raise ValueError("train/val/test ratios must sum to 1.0")

    by_speaker: dict[str, list[dict]] = defaultdict(list)
    for record in records:
        by_speaker[record["speaker_id"]].append(record)

    speakers = list(by_speaker.keys())
    rng = random.Random(seed)
    rng.shuffle(speakers)

    n = len(speakers)
```

```
    n_train = int(n * train_ratio)
    n_val = int(n * val_ratio)

    train_speakers = set(speakers[:n_train])
    val_speakers = set(speakers[n_train : n_train + n_val])
    test_speakers = set(speakers[n_train + n_val :])

    assert not (train_speakers & val_speakers)
    assert not (train_speakers & test_speakers)
    assert not (val_speakers & test_speakers)

    splits: dict[SplitName, list[dict]] = {"train": [], "val": [],
"test": []}
    for speaker_id, speaker_records in by_speaker.items():
        if speaker_id in train_speakers:
            split: SplitName = "train"
        elif speaker_id in val_speakers:
            split = "val"
        else:
            split = "test"
        for record in speaker_records:
            row = dict(record)
            row["split"] = split
            splits[split].append(row)

    return splits


def utterance_random_split(
    records: list[dict],
    *,
    seed: int = DEFAULT_SEED,
    train_ratio: float = TRAIN_RATIO,
    val_ratio: float = VAL_RATIO,
    test_ratio: float = TEST_RATIO,
) -> dict[SplitName, list[dict]]:
    """"Leaky" split: shuffle utterances directly, ignoring speaker
identity.

    Same size/ratios as speaker_disjoint_split, but a speaker's
utterances can
    land in multiple splits. Used only as a Phase 9 ablation to
measure how
    much speaker leakage inflates apparent performance on this
dataset.
    """
    if abs(train_ratio + val_ratio + test_ratio - 1.0) > 1e-6:
        raise ValueError("train/val/test ratios must sum to 1.0")

    shuffled = list(records)
```

```
    random.Random(seed).shuffle(shuffled)

    n = len(shuffled)
    n_train = int(n * train_ratio)
    n_val = int(n * val_ratio)

    splits: dict[SplitName, list[dict]] = {"train": [], "val": [],
"test": []}
    for i, record in enumerate(shuffled):
        row = dict(record)
        if i < n_train:
            row["split"] = "train"
            splits["train"].append(row)
        elif i < n_train + n_val:
            row["split"] = "val"
            splits["val"].append(row)
        else:
            row["split"] = "test"
            splits["test"].append(row)
    return splits


def count_speaker_overlap(splits: dict[SplitName, list[dict]]) -> int:
    """Return number of speakers appearing in more than one split
(expect 0)."""
    speaker_to_splits: dict[str, set[str]] = defaultdict(set)
    for split_name, rows in splits.items():
        for row in rows:
            speaker_to_splits[row["speaker_id"]].add(split_name)
    return sum(1 for splits_seen in speaker_to_splits.values() if
len(splits_seen) > 1)
```

src/wer_metrics.py

```
"""Minimal, model-agnostic WER/CER computation (jiwer-based)."""
from __future__ import annotations

import jiwer


def compute_wer_cer(references: list[str], hypotheses: list[str]) ->
tuple[float, float]:
    wer = jiwer.wer(references, hypotheses)
    cer = jiwer.cer(references, hypotheses)
    return wer, cer
```

src/xlsr_training.py

```
"""XLS-R (wav2vec2) CTC fine-tuning on the speaker-disjoint OpenSLR-54
subset.
```

```
Phase 6: fine-tune the audited checkpoint (gagan3012/wav2vec2-xlsr-
nepali)
on speaker-diverse data to close its demonstrated generalization gap.
Keeps the checkpoint's own tokenizer/vocab (we repair the same model,
not
train a new one). Experiment tracking: MLflow (report_to=["mlflow"]).

Collator adapted from the Hugging Face wav2vec2 fine-tuning blog
(https://huggingface.co/blog/fine-tune-wav2vec2-english) -- cite in
the report.
"""
from __future__ import annotations

from dataclasses import dataclass
from pathlib import Path
from typing import Any

import numpy as np
import torch
from datasets import Audio, Dataset
from transformers import (
    EarlyStoppingCallback,
    Trainer,
    TrainerCallback,
    TrainingArguments,
    Wav2Vec2ForCTC,
    Wav2Vec2Processor,
    set_seed,
)


class MPSCacheClearCallback(TrainerCallback):
    """Long MPS training runs accumulate allocator fragmentation until
    torch.autograd's backward pass OOMs (observed: crashed ~1580 steps
in,
    17GB allocated, right after an eval pass). Periodic empty_cache()
keeps
    the allocator from growing unbounded. No-op on CUDA/CPU."""

    def _clear(self) -> None:
        if torch.backends.mps.is_available():
            torch.mps.empty_cache()

    def on_step_end(self, args, state, control, **kwargs):
        if state.global_step % 50 == 0:
            self._clear()

    def on_evaluate(self, args, state, control, **kwargs):
        self._clear()
```

```python
from src.manifests import read_jsonl_manifest
from src.wer_metrics import compute_wer_cer
from src.xlsr_baseline import XLSR_MODEL_ID, clean_transcript

DEFAULT_SEED = 42
TARGET_SAMPLE_RATE = 16_000
# The checkpoint is already Nepali fine-tuned; a low LR adapts it to
diverse
# speakers without erasing what it knows (Phase 7 checks forgetting
anyway).
DEFAULT_LEARNING_RATE = 3e-5
MAX_EPOCHS = 5


def load_model_and_processor(
    model_id: str = XLSR_MODEL_ID,
) -> tuple[Wav2Vec2ForCTC, Wav2Vec2Processor]:
    processor = Wav2Vec2Processor.from_pretrained(model_id)
    # torch's scaled_dot_product_attention raises NotImplementedError
on Apple
    # MPS when dropout is active (i.e. in training mode); fall back to
eager
    # attention off-CUDA so local smoke tests run. CUDA (Colab T4)
keeps SDPA.
    attn = "sdpa" if torch.cuda.is_available() else "eager"
    model = Wav2Vec2ForCTC.from_pretrained(model_id,
attn_implementation=attn)
    # Standard wav2vec2 fine-tuning practice: the convolutional
feature
    # encoder was trained on far more audio than we have -- freeze it.
    model.freeze_feature_encoder()
    return model, processor


def manifest_to_dataset(rows: list[dict], project_root: Path) ->
Dataset:
    def _generator() -> Any:
        for row in rows:
            yield {
                "audio": str((project_root /
row["audio_path"]).resolve()),
                "text": clean_transcript(row["transcript"]),
                "utterance_id": row["utterance_id"],
            }

    dataset = Dataset.from_generator(_generator)
    return dataset.cast_column("audio",
Audio(sampling_rate=TARGET_SAMPLE_RATE))
```

```
def load_datasets(
    project_root: Path,
    *,
    manifest_dir: Path | None = None,
    max_train: int | None = None,
    max_eval: int | None = None,
) -> tuple[Dataset, Dataset]:
    manifest_dir = manifest_dir or (project_root / "data" /
"manifests")
    train_rows = read_jsonl_manifest(manifest_dir / "train.jsonl")
    val_rows = read_jsonl_manifest(manifest_dir / "val.jsonl")
    if max_train is not None:
        train_rows = train_rows[:max_train]
    if max_eval is not None:
        val_rows = val_rows[:max_eval]
    return (
        manifest_to_dataset(train_rows, project_root),
        manifest_to_dataset(val_rows, project_root),
    )


def build_prepare_fn(processor: Wav2Vec2Processor):
    def prepare(batch: dict) -> dict:
        audio = batch["audio"]
        batch["input_values"] = processor(
            audio["array"], sampling_rate=audio["sampling_rate"]
        ).input_values[0]
        batch["labels"] = processor.tokenizer(batch["text"]).input_ids
        return batch

    return prepare


@dataclass
class DataCollatorCTCWithPadding:
    processor: Wav2Vec2Processor

    def __call__(self, features: list[dict]) -> dict[str,
torch.Tensor]:
        input_features = [{"input_values": f["input_values"]} for f in
features]
        label_features = [{"input_ids": f["labels"]} for f in
features]

        batch = self.processor.feature_extractor.pad(input_features,
return_tensors="pt")
        labels_batch = self.processor.tokenizer.pad(label_features,
return_tensors="pt")
```

```
        labels = labels_batch["input_ids"].masked_fill(
            labels_batch.attention_mask.ne(1), -100
        )
        batch["labels"] = labels
        return batch


def build_compute_metrics(processor: Wav2Vec2Processor):
    def compute_metrics(pred) -> dict[str, float]:
        pred_ids = np.argmax(pred.predictions, axis=-1)
        label_ids = np.where(
            pred.label_ids != -100, pred.label_ids,
processor.tokenizer.pad_token_id
        )
        pred_str = processor.batch_decode(pred_ids)
        label_str = processor.batch_decode(label_ids,
group_tokens=False)
        wer, cer = compute_wer_cer(label_str, pred_str)
        return {"wer": wer, "cer": cer}

    return compute_metrics


def build_training_arguments(
    output_dir: Path,
    *,
    smoke_test: bool = False,
    learning_rate: float = DEFAULT_LEARNING_RATE,
) -> TrainingArguments:
    fp16 = torch.cuda.is_available()
    if smoke_test:
        return TrainingArguments(
            output_dir=str(output_dir),
            per_device_train_batch_size=2,
            per_device_eval_batch_size=2,
            gradient_accumulation_steps=4,
            learning_rate=learning_rate,
            warmup_steps=0,
            max_steps=6,
            eval_strategy="steps",
            eval_steps=3,
            logging_steps=1,
            save_steps=3,
            save_total_limit=2,
            fp16=fp16,
            report_to=["mlflow"],
            remove_unused_columns=False,
            label_names=["labels"],
            load_best_model_at_end=False,
```

```
        )

    return TrainingArguments(
        output_dir=str(output_dir),
        per_device_train_batch_size=2,
        per_device_eval_batch_size=2,
        gradient_accumulation_steps=4,
        learning_rate=learning_rate,
        warmup_steps=500,
        num_train_epochs=MAX_EPOCHS,
        eval_strategy="epoch",
        save_strategy="epoch",
        logging_steps=100,
        save_total_limit=3,
        fp16=fp16,
        report_to=["mlflow"],
        remove_unused_columns=False,
        label_names=["labels"],
        load_best_model_at_end=True,
        metric_for_best_model="wer",
        greater_is_better=False,
    )


def create_trainer(
    project_root: Path,
    output_dir: Path,
    *,
    smoke_test: bool = False,
    seed: int = DEFAULT_SEED,
    manifest_dir: Path | None = None,
    max_train: int | None = None,
    max_eval: int | None = None,
) -> tuple[Trainer, Wav2Vec2Processor]:
    set_seed(seed)
    model, processor = load_model_and_processor()

    if smoke_test and max_train is None:
        max_train, max_eval = 32, 8
    train_ds, eval_ds = load_datasets(
        project_root, manifest_dir=manifest_dir, max_train=max_train,
max_eval=max_eval
    )

    prepare = build_prepare_fn(processor)
    train_ds = train_ds.map(prepare,
remove_columns=train_ds.column_names)
    eval_ds = eval_ds.map(prepare,
remove_columns=eval_ds.column_names)
```

```
    args = build_training_arguments(output_dir, smoke_test=smoke_test)
    trainer = Trainer(
        model=model,
        args=args,
        train_dataset=train_ds,
        eval_dataset=eval_ds,
        data_collator=DataCollatorCTCWithPadding(processor=processor),
        compute_metrics=build_compute_metrics(processor),
        processing_class=processor.feature_extractor,
    )
    trainer.add_callback(MPSCacheClearCallback())
    if not smoke_test:

trainer.add_callback(EarlyStoppingCallback(early_stopping_patience=2))
    return trainer, processor


def train_and_save(
    project_root: Path,
    output_dir: Path,
    *,
    smoke_test: bool = False,
    resume_from_checkpoint: str | bool | None = None,
    seed: int = DEFAULT_SEED,
    manifest_dir: Path | None = None,
    max_train: int | None = None,
    max_eval: int | None = None,
) -> dict:
    trainer, processor = create_trainer(
        project_root,
        output_dir,
        smoke_test=smoke_test,
        seed=seed,
        manifest_dir=manifest_dir,
        max_train=max_train,
        max_eval=max_eval,
    )
    train_result =
trainer.train(resume_from_checkpoint=resume_from_checkpoint)
    eval_metrics = trainer.evaluate()
    trainer.save_model(str(output_dir))
    processor.save_pretrained(str(output_dir))
    return {
        "train_loss": train_result.training_loss,
        "eval_metrics": eval_metrics,
        "checkpoint_dir": str(output_dir),
        "global_step": trainer.state.global_step,
    }
```

scripts/make_leaky_manifests.py

```python
"""Phase 9 ablation: build a leaky (utterance-random, non-speaker-
disjoint)
split from the same processed records as the real manifests, for a
matched-training-budget comparison against the speaker-disjoint
split."""
from __future__ import annotations

import sys
from pathlib import Path

sys.path.insert(0, str(Path(__file__).resolve().parents[1]))

from src.manifests import read_jsonl_manifest, write_split_manifests
from src.splits import count_speaker_overlap, utterance_random_split

PROJECT_ROOT = Path(__file__).resolve().parents[1]


def main() -> None:
    all_rows = read_jsonl_manifest(PROJECT_ROOT / "data" / "manifests"
/ "all.jsonl")
    splits = utterance_random_split(all_rows, seed=42)
    manifest_dir = PROJECT_ROOT / "data" / "manifests_leaky"
    write_split_manifests(splits, manifest_dir)
    overlap = count_speaker_overlap(splits)
    print(f"Leaky split written to {manifest_dir}")
    print(f"train={len(splits['train'])} val={len(splits['val'])}
test={len(splits['test'])}")
    print(f"speaker_overlap between splits: {overlap} (expected > 0 --
this is the leaky ablation)")


if __name__ == "__main__":
    main()
```

scripts/run_efficiency_benchmark.py

```python
"""Phase 8 — Efficiency / Deployment Benchmark.

Benchmarks the fine-tuned checkpoint (models/xlsr-ft): model size, CPU
inference latency, and the FP32 vs. dynamic int8 quantization trade-
off
(latency speedup vs. WER/CER cost). Logged to MLflow (experiment
"phase8-efficiency").
"""
from __future__ import annotations

import json
import random
```

```python
import sys
import time
from pathlib import Path

sys.path.insert(0, str(Path(__file__).resolve().parents[1]))

import mlflow
import soundfile as sf
import torch
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor

# Apple Silicon (and most non-x86 ARM) only ships the qnnpack
quantized
# backend, not fbgemm (torch's x86 default) -- without this,
quantize_dynamic
# raises "RuntimeError: Didn't find engine for operation ...
NoQEngine".
if "qnnpack" in torch.backends.quantized.supported_engines:
    torch.backends.quantized.engine = "qnnpack"

from src.manifests import read_jsonl_manifest
from src.wer_metrics import compute_wer_cer
from src.xlsr_baseline import resample_if_needed

PROJECT_ROOT = Path(__file__).resolve().parents[1]
CHECKPOINT_DIR = PROJECT_ROOT / "models" / "xlsr-ft"
N_SAMPLES = 50
N_LATENCY_RUNS = 20


def model_size_mb(model: torch.nn.Module) -> float:
    total_bytes = sum(p.numel() * p.element_size() for p in
model.parameters())
    total_bytes += sum(b.numel() * b.element_size() for b in
model.buffers())
    return total_bytes / (1024 * 1024)


def transcribe(model, processor, speech, device: str) -> str:
    inputs = processor(speech, sampling_rate=16_000,
return_tensors="pt")
    inputs = {k: v.to(device) for k, v in inputs.items()}
    with torch.no_grad():
        logits = model(**inputs).logits
    pred_ids = torch.argmax(logits, dim=-1)
    return processor.batch_decode(pred_ids)[0]


def benchmark_latency(model, processor, samples: list, device: str) ->
```

```
dict:
    # Warm-up (first call pays one-time graph/cache setup cost)
    transcribe(model, processor, samples[0], device)
    times = []
    for speech in samples[:N_LATENCY_RUNS]:
        start = time.perf_counter()
        transcribe(model, processor, speech, device)
        times.append(time.perf_counter() - start)
    return {
        "mean_latency_s": sum(times) / len(times),
        "min_latency_s": min(times),
        "max_latency_s": max(times),
        "num_runs": len(times),
    }


def evaluate_wer(model, processor, rows: list, device: str) -> dict:
    refs, hyps = [], []
    for row in rows:
        audio, sr = sf.read(PROJECT_ROOT / row["audio_path"])
        speech = resample_if_needed(audio, sr)
        hyps.append(transcribe(model, processor, speech, device))
        refs.append(row["transcript"])
    wer, cer = compute_wer_cer(refs, hyps)
    return {"wer": wer, "cer": cer, "num_utterances": len(rows)}


def main() -> None:
    device = "cpu"  # deployment-realistic target; dynamic
quantization is CPU-only anyway
    processor = Wav2Vec2Processor.from_pretrained(CHECKPOINT_DIR)

    rows = read_jsonl_manifest(PROJECT_ROOT / "data" / "manifests" /
"test.jsonl")
    shuffled = list(rows)
    random.Random(42).shuffle(shuffled)
    sample_rows = shuffled[:N_SAMPLES]
    samples = []
    for row in sample_rows:
        audio, sr = sf.read(PROJECT_ROOT / row["audio_path"])
        samples.append(resample_if_needed(audio, sr))

    print("=== FP32 ===")
    model_fp32 = Wav2Vec2ForCTC.from_pretrained(CHECKPOINT_DIR)
    model_fp32.to(device).eval()
    fp32_size = model_size_mb(model_fp32)
    fp32_latency = benchmark_latency(model_fp32, processor, samples,
device)
    fp32_wer = evaluate_wer(model_fp32, processor, sample_rows,
device)
```

```
    print(f"size={fp32_size:.1f}MB
latency={fp32_latency['mean_latency_s']*1000:.1f}ms
wer={fp32_wer['wer']:.4f}")

    print("=== Dynamic INT8 (torch.quantization.quantize_dynamic on
Linear layers) ===")
    model_int8 = torch.quantization.quantize_dynamic(
        model_fp32, {torch.nn.Linear}, dtype=torch.qint8
    )
    int8_size = model_size_mb(model_int8)
    int8_latency = benchmark_latency(model_int8, processor, samples,
device)
    int8_wer = evaluate_wer(model_int8, processor, sample_rows,
device)
    print(f"size={int8_size:.1f}MB
latency={int8_latency['mean_latency_s']*1000:.1f}ms
wer={int8_wer['wer']:.4f}")

    results = {
        "checkpoint": str(CHECKPOINT_DIR),
        "n_samples": N_SAMPLES,
        "n_latency_runs": N_LATENCY_RUNS,
        "device": "cpu",
        "fp32": {"model_size_mb": fp32_size, **fp32_latency,
**fp32_wer},
        "int8_dynamic": {"model_size_mb": int8_size, **int8_latency,
**int8_wer},
        "speedup_x": fp32_latency["mean_latency_s"] /
int8_latency["mean_latency_s"],
        "size_reduction_pct": (1 - int8_size / fp32_size) * 100,
        "wer_delta": int8_wer["wer"] - fp32_wer["wer"],
    }

    mlflow.set_tracking_uri(f"file://{PROJECT_ROOT / 'mlruns'}")
    mlflow.set_experiment("phase8-efficiency")
    with mlflow.start_run(run_name="fp32_vs_int8"):
        mlflow.log_params({"checkpoint": str(CHECKPOINT_DIR),
"n_samples": N_SAMPLES})
        mlflow.log_metrics(
            {
                "fp32_size_mb": fp32_size,
                "fp32_latency_ms": fp32_latency["mean_latency_s"] *
1000,
                "fp32_wer": fp32_wer["wer"],
                "int8_size_mb": int8_size,
                "int8_latency_ms": int8_latency["mean_latency_s"] *
1000,
                "int8_wer": int8_wer["wer"],
                "speedup_x": results["speedup_x"],
            }
```

```python
        )

    reports_dir = PROJECT_ROOT / "reports"
    reports_dir.mkdir(parents=True, exist_ok=True)
    out = reports_dir / "phase8_efficiency_results.json"
    out.write_text(json.dumps(results, indent=2), encoding="utf-8")
    print(f"\nSpeedup: {results['speedup_x']:.2f}x, size reduction:
{results['size_reduction_pct']:.1f}%, WER delta:
{results['wer_delta']*100:+.2f}pp")
    print(f"Saved: {out}")


if __name__ == "__main__":
    main()
```

scripts/run_generalization_eval.py

```python
"""Phase 7 — Generalization Re-Evaluation.

Evaluates a checkpoint (original audited model, OR the Phase 6 fine-
tuned
one) on BOTH test sets, producing the four-cell comparison table:
    {original, fine-tuned} x {in-domain OpenSLR-43, out-of-domain
OpenSLR-54}

Run once per checkpoint, then diff the two resulting JSON files:
    python scripts/run_generalization_eval.py --checkpoint
gagan3012/wav2vec2-xlsr-nepali --tag original
    python scripts/run_generalization_eval.py --checkpoint
models/xlsr-ft --tag finetuned

All runs logged to MLflow (experiment "phase7-generalization").
"""
from __future__ import annotations

import argparse
import json
import random
import sys
from pathlib import Path

sys.path.insert(0, str(Path(__file__).resolve().parents[1]))

import mlflow
import soundfile as sf
import torch
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor

from src.manifests import read_jsonl_manifest
from src.wer_metrics import compute_wer_cer
```

```python
from src.xlsr_baseline import (
    build_openslr43_manifest_rows,
    load_openslr43_test_dataset,
    resample_if_needed,
    transcribe_ctc,
)

PROJECT_ROOT = Path(__file__).resolve().parents[1]
REPORTS_DIR = PROJECT_ROOT / "reports"


def eval_slr43(model, processor, device: str, n_samples: int) -> dict:
    dataset = load_openslr43_test_dataset(max_samples=n_samples)
    rows = build_openslr43_manifest_rows(dataset)
    refs, hyps = [], []
    for i, row in enumerate(dataset):
        audio = row["audio"]
        speech = resample_if_needed(audio["array"],
audio["sampling_rate"])
        hyps.append(transcribe_ctc(model, processor, speech, device))
        refs.append(rows[i]["text"])
        if (i + 1) % 25 == 0:
            print(f"  [SLR43] {i + 1}/{len(dataset)}")
    wer, cer = compute_wer_cer(refs, hyps)
    return {"dataset": "OpenSLR-43 (in-domain)", "num_utterances":
len(dataset), "wer": wer, "cer": cer}


def eval_slr54(model, processor, device: str, n_samples: int, seed:
int) -> dict:
    all_rows = read_jsonl_manifest(PROJECT_ROOT / "data" / "manifests"
/ "test.jsonl")
    shuffled = list(all_rows)
    random.Random(seed).shuffle(shuffled)
    rows = shuffled[:n_samples] if n_samples else shuffled

    refs, hyps = [], []
    for i, row in enumerate(rows):
        audio, sr = sf.read(PROJECT_ROOT / row["audio_path"])
        speech = resample_if_needed(audio, sr)
        hyps.append(transcribe_ctc(model, processor, speech, device))
        refs.append(row["transcript"])
        if (i + 1) % 25 == 0:
            print(f"  [SLR54] {i + 1}/{len(rows)}")
    wer, cer = compute_wer_cer(refs, hyps)
    return {"dataset": "OpenSLR-54 speaker-disjoint test (out-of-
domain)", "num_utterances": len(rows), "wer": wer, "cer": cer}


def main() -> None:
```

```
    parser = argparse.ArgumentParser()
    parser.add_argument("--checkpoint", required=True, help="HF model
id or local path")
    parser.add_argument("--tag", required=True, help="'original' or
'finetuned' (used in filenames/MLflow run names)")
    parser.add_argument("--n_samples", type=int, default=150)
    parser.add_argument("--seed", type=int, default=42)
    args = parser.parse_args()

    device = "cuda" if torch.cuda.is_available() else "cpu"
    print(f"Loading {args.checkpoint} ({args.tag})...")
    processor = Wav2Vec2Processor.from_pretrained(args.checkpoint)
    model = Wav2Vec2ForCTC.from_pretrained(args.checkpoint)
    model.to(device)
    model.eval()

    mlflow.set_tracking_uri(f"file://{PROJECT_ROOT / 'mlruns'}")
    mlflow.set_experiment("phase7-generalization")

    slr43 = eval_slr43(model, processor, device, args.n_samples)
    slr54 = eval_slr54(model, processor, device, args.n_samples,
args.seed)
    print(f"[{args.tag}] SLR43 in-domain:     WER={slr43['wer']:.4f}
CER={slr43['cer']:.4f}")
    print(f"[{args.tag}] SLR54 out-of-domain: WER={slr54['wer']:.4f}
CER={slr54['cer']:.4f}")

    with mlflow.start_run(run_name=f"{args.tag}_slr43"):
        mlflow.log_params({"checkpoint": args.checkpoint, "tag":
args.tag, "dataset": "slr43"})
        mlflow.log_metrics({"wer": slr43["wer"], "cer": slr43["cer"]})
    with mlflow.start_run(run_name=f"{args.tag}_slr54"):
        mlflow.log_params({"checkpoint": args.checkpoint, "tag":
args.tag, "dataset": "slr54"})
        mlflow.log_metrics({"wer": slr54["wer"], "cer": slr54["cer"]})

    REPORTS_DIR.mkdir(parents=True, exist_ok=True)
    out = REPORTS_DIR / f"phase7_{args.tag}_results.json"
    out.write_text(
        json.dumps({"checkpoint": args.checkpoint, "tag": args.tag,
"in_domain_openslr43": slr43, "out_of_domain_openslr54": slr54},
indent=2),
        encoding="utf-8",
    )
    print(f"Saved: {out}")

    other_tag = "finetuned" if args.tag == "original" else "original"
    other_path = REPORTS_DIR / f"phase7_{other_tag}_results.json"
    if other_path.exists():
        other = json.loads(other_path.read_text())
```

```
        print("\n=== Four-cell comparison ===")
        print(f"{'':12} {'in-domain (SLR43)':>20} {'out-of-domain
(SLR54)':>24}")
        a, b = (out, other_path) if args.tag == "original" else
(other_path, out)
        a_data, b_data = json.loads(a.read_text()),
json.loads(b.read_text())
        print(f"{'original':12} {a_data['in_domain_openslr43']
['wer']*100:>18.2f}% {a_data['out_of_domain_openslr54']
['wer']*100:>22.2f}%")
        print(f"{'finetuned':12} {b_data['in_domain_openslr43']
['wer']*100:>18.2f}% {b_data['out_of_domain_openslr54']
['wer']*100:>22.2f}%")


if __name__ == "__main__":
    main()
```

Appendix C — Evidence and Reproducibility Artifacts

This section indexes the reproducibility evidence for this project. Screenshots referenced below are inserted as images where available; all underlying raw data is additionally available as tracked files in the project repository (`coursework-10phase` branch) for direct inspection, per the reproducibility requirement.

C.1 Environment

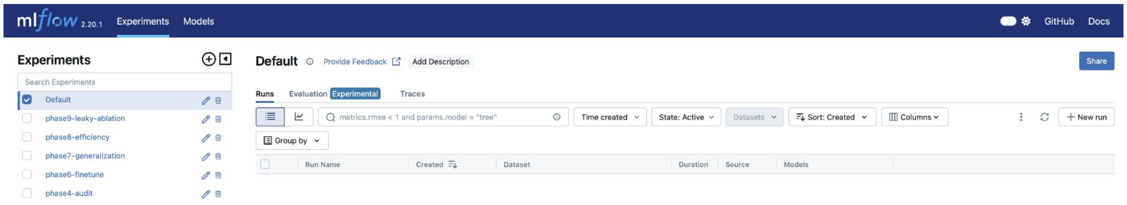
```
=== Environment Info ===
Platform: macOS-26.5.2-arm64-arm-64bit
Python: 3.11.15 | packaged by conda-forge | (main, Jun 11 2026,
03:29:05) [Clang 19.1.7 ]
torch: 2.13.0
CUDA available: False
MPS available: True
transformers: 4.49.0
datasets: 3.2.0
mlflow: 2.20.1
```

```
cat /Users/bimochankunwar/Desktop/NSTT-Lite/appendix_screenshots/environ
ment_info.txt
(base) bimochankunwar@Bimochans-MacBook-Pro NSTT-Lite % cat /Users/bimoc
ha
nkunwar/Desktop/NSTT-Lite/appendix_sc
reenshots/environment_info.txt
=== Environment Info ===
Platform: macOS-26.5.2-arm64-arm-64bit
Python: 3.11.15 | packaged by conda-forge | (main, Jun 11 2026, 03:29:05
) [Clang 19.1.7 ]
torch: 2.13.0
CUDA available: False
MPS available: True
transformers: 4.49.0
datasets: 3.2.0
mlflow: 2.20.1
(base) bimochankunwar@Bimochans-MacBook-Pro NSTT-Lite %
```

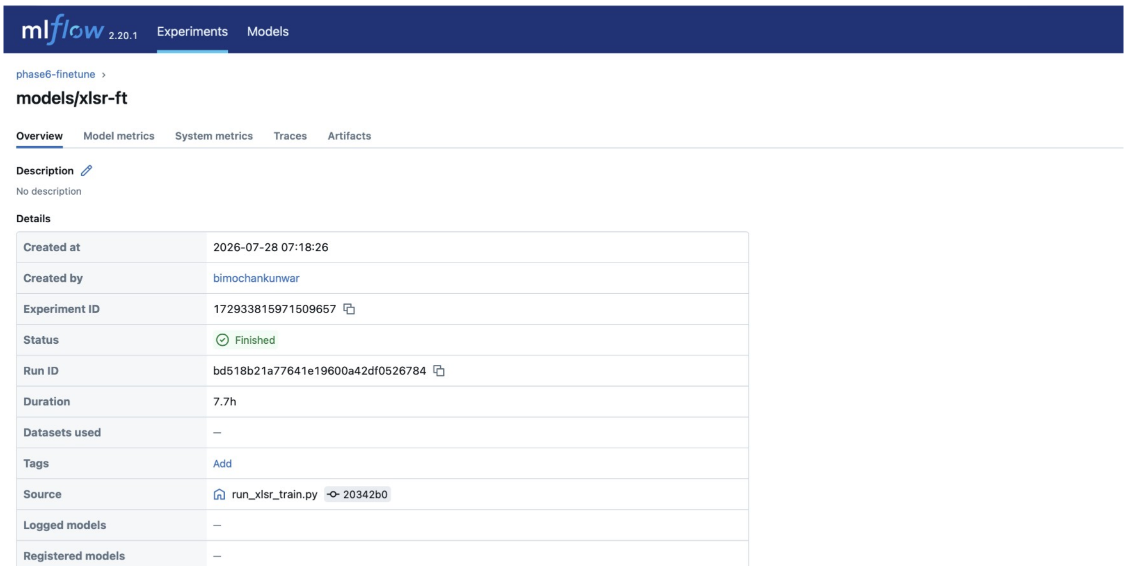
Terminal output of the environment info script

C.2 MLflow experiment tracking

Six MLflow experiments were logged over the course of this project (local `mlruns/` store): `phase4-audit`, `phase6-finetune`, `phase7-generalization`, `phase8-efficiency`, `phase9-leaky-ablation`, plus the default experiment used during early smoke testing. Each run logs its parameters, metrics, and (where applicable) example artifacts.



MLflow experiments list



Run overview for the full training run (Run ID bd518b21…, Duration 7.7h, Status Finished, Source run_xlsr_train.py)

Model metrics: eval_wer curve (0.7098 → 0.6883 across the last four logged epochs) and train_loss (0.50)

(Additional run and metric screenshots for all six experiments are available in the project repository's `appendix_screenshots/` folder and the tracked `mlruns/` store, per the reproducibility note above; only representative examples are shown here.)

Appendix D — Extended Results Tables

Full machine-readable result files for every phase-4/7/8/9 measurement quoted in the report, plus the complete per-utterance error-analysis listing (150 utterances, Section 4.4), for independent verification beyond the summary tables in the main report body.

D.1 Summary result files

`reports/phase4_audit_results.json`


```json
{
  "model": "gagan3012/wav2vec2-xlsr-nepali",
  "self_reported_wer": 0.0597,
  "device": {
    "platform": "macOS-26.5.2-arm64-arm-64bit-Mach-O",
    "python": "3.13.13",
    "torch": "2.13.0",
    "cuda_available": false,
    "mps_available": true
  },
  "in_domain_openslr43": {
    "dataset": "OpenSLR-43 (in-domain)",
    "num_utterances": 150,
    "wer": 0.04913494809688582,
    "cer": 0.008724212254952273
```

```
  },
  "out_of_domain_openslr54": {
    "dataset": "OpenSLR-54 speaker-disjoint test (out-of-domain)",
    "num_utterances": 150,
    "wer": 0.6229508196721312,
    "cer": 0.17383314957736126
  },
  "corpus_separation_note": "OpenSLR-43 and OpenSLR-54 results are
never merged into a single number.",
  "slr54_gender_note": "SLR54 rows carry an F0-pitch gender pseudo-
label (165Hz threshold), not verified metadata -- used only for Phase
9 per-group analysis."
}
```

reports/phase7_original_results.json

```
{
  "checkpoint": "gagan3012/wav2vec2-xlsr-nepali",
  "tag": "original",
  "in_domain_openslr43": {
    "dataset": "OpenSLR-43 (in-domain)",
    "num_utterances": 150,
    "wer": 0.04913494809688582,
    "cer": 0.008724212254952273
  },
  "out_of_domain_openslr54": {
    "dataset": "OpenSLR-54 speaker-disjoint test (out-of-domain)",
    "num_utterances": 150,
    "wer": 0.6229508196721312,
    "cer": 0.17383314957736126
  }
}
```

reports/phase7_finetuned_results.json

```
{
  "checkpoint": "models/xlsr-ft",
  "tag": "finetuned",
  "in_domain_openslr43": {
    "dataset": "OpenSLR-43 (in-domain)",
    "num_utterances": 150,
    "wer": 0.16401384083044981,
    "cer": 0.029764959458072462
  },
  "out_of_domain_openslr54": {
    "dataset": "OpenSLR-54 speaker-disjoint test (out-of-domain)",
    "num_utterances": 150,
    "wer": 0.38173302107728335,
    "cer": 0.09886071297317163
  }
}
```

reports/phase7_leaky_results.json


```json
{
  "checkpoint": "models/xlsr-ft-leaky",
  "tag": "leaky",
  "in_domain_openslr43": {
    "dataset": "OpenSLR-43 (in-domain)",
    "num_utterances": 150,
    "wer": 0.1695501730103806,
    "cer": 0.03048342399671559
  },
  "out_of_domain_openslr54": {
    "dataset": "OpenSLR-54 speaker-disjoint test (out-of-domain)",
    "num_utterances": 150,
    "wer": 0.3255269320843091,
    "cer": 0.08195516354281514
  }
}
```


reports/phase8_efficiency_results.json


```json
{
  "checkpoint": "/Users/bimochankunwar/Desktop/NSTT-Lite/models/xlsr-
ft",
  "n_samples": 50,
  "n_latency_runs": 20,
  "device": "cpu",
  "fp32": {
    "model_size_mb": 1203.5573768615723,
    "mean_latency_s": 0.20469055199064315,
    "min_latency_s": 0.14858783304225653,
    "max_latency_s": 0.3167113340459764,
    "num_runs": 20,
    "wer": 0.3581081081081081,
    "cer": 0.07165775401069518,
    "num_utterances": 50
  },
  "int8_dynamic": {
    "model_size_mb": 48.45556640625,
    "mean_latency_s": 0.33524910402484237,
    "min_latency_s": 0.20782845804933459,
    "max_latency_s": 0.5563514999812469,
    "num_runs": 20,
    "wer": 0.36486486486486486,
    "cer": 0.0748663101604278,
    "num_utterances": 50
  },
  "speedup_x": 0.6105625623848806,
  "size_reduction_pct": 95.97397121750821,
  "wer_delta": 0.006756756756756743
}
```

reports/phase9_error_analysis_results.json


```json
{
  "checkpoint": "/Users/bimochankunwar/Desktop/NSTT-Lite/models/xlsr-
ft",
  "num_utterances": 150,
  "overall_wer": 0.38173302107728335,
  "overall_cer": 0.09886071297317163,
  "error_category_counts": {
    "other": 59,
    "oov_rare_vocabulary": 90,
    "phonetic_confusion": 64,
    "noise_degradation": 22
  },
  "error_category_note": "Heuristic, regex/jiwer-alignment-based
tagging -- categories can overlap per utterance and are not mutually
exclusive.",
  "gender_pseudo_label_breakdown": {
    "male": {
      "wer": 0.37668161434977576,
      "cer": 0.10310734463276836,
      "num_utterances": 77
    },
    "female": {
      "wer": 0.3872549019607843,
      "cer": 0.09425287356321839,
      "num_utterances": 73
    }
  },
  "gender_pseudo_label_note": "gender_pseudo_label is an F0-pitch
threshold heuristic (165Hz), NOT verified ground truth -- this
breakdown measures WER by acoustic pitch-threshold group, not by true
gender.",
  "speaker_wer_spread": {
    "min": [
      "76dab",
      {
        "mean_wer": 0.125,
        "n": 12
      }
    ],
    "max": [
      "70e10",
      {
        "mean_wer": 0.6363636363636364,
        "n": 11
      }
    ],
    "num_speakers": 16
  },
  "worst_10_examples": [
```

```
{
  "utterance_id": "8c693c9a97",
  "speaker_id": "7f9c6",
  "gender_pseudo_label": "female",
  "reference": "मान्छे चटकारे",
  "hypothesis": "मान्छि चट कार्य",
  "wer": 1.5,
  "cer": 0.3076923076923077,
  "categories": [
    "noise_degradation",
    "oov_rare_vocabulary",
    "phonetic_confusion"
  ]
},
{
  "utterance_id": "d443c84109",
  "speaker_id": "70e10",
  "gender_pseudo_label": "female",
  "reference": "क्रान्तिकारी वाममोर्चाको",
  "hypothesis": "तन्थिकारी बाम वर्षको",
  "wer": 1.5,
  "cer": 0.4166666666666667,
  "categories": [
    "noise_degradation",
    "oov_rare_vocabulary",
    "phonetic_confusion"
  ]
},
{
  "utterance_id": "7da0e9a7b7",
  "speaker_id": "99866",
  "gender_pseudo_label": "female",
  "reference": "राज्यमाथि हासिल गऱ्यो",
  "hypothesis": "राज्यमाति हासेल गरयो",
  "wer": 1.0,
  "cer": 0.13636363636363635,
  "categories": [
    "noise_degradation",
    "oov_rare_vocabulary",
    "phonetic_confusion"
  ]
},
{
  "utterance_id": "8ccd499b56",
  "speaker_id": "da0cf",
  "gender_pseudo_label": "male",
```

    "reference": "बनाउनबाट जोगाउँदछ",
    "hypothesis": "बनाउनबाटा जोगआउँदछ",
    "wer": 1.0,
    "cer": 0.11764705882352941,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "0cf3ebaf89",
    "speaker_id": "07179",
    "gender_pseudo_label": "female",
    "reference": "उद्देश्यले ५ अप्रिल",
    "hypothesis": "उद्येश्यले पच्च अपरेल",
    "wer": 1.0,
    "cer": 0.2631578947368421,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "3cd1567940",
    "speaker_id": "e8f6f",
    "gender_pseudo_label": "male",
    "reference": "ब्याज तिर्नुपर्ने हुन्छ",
    "hypothesis": "ब्यहास दिन्नु पर्ने हुन्छ",
    "wer": 1.0,
    "cer": 0.2608695652173913,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "0950a3e9a4",
    "speaker_id": "99866",
    "gender_pseudo_label": "female",
    "reference": "रन्मामैकोट तकसेरा हुकाम",
    "hypothesis": "रणमा मैकोट तक्सेरा हुकाम",
    "wer": 1.0,

```
    "cer": 0.17391304347826086,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "7b1f434978",
    "speaker_id": "99866",
    "gender_pseudo_label": "female",
    "reference": "आइल्यान्ड तथा बेलायतदेखि",
    "hypothesis": "आइल्यान् तथा बेलाय देखि",
    "wer": 1.0,
    "cer": 0.08333333333333333,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "34760118f4",
    "speaker_id": "9d08c",
    "gender_pseudo_label": "male",
    "reference": "वृक्ष यही हो",
    "hypothesis": "बृच्ष यहियो",
    "wer": 1.0,
    "cer": 0.4166666666666667,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
    ]
  },
  {
    "utterance_id": "dde677080a",
    "speaker_id": "70e10",
    "gender_pseudo_label": "female",
    "reference": "सूर्यले पृथ्वीको वरिपरि",
    "hypothesis": "सवर्यलले फृथ्ष गरीपुरि",
    "wer": 1.0,
    "cer": 0.43478260869565216,
    "categories": [
      "noise_degradation",
      "oov_rare_vocabulary",
      "phonetic_confusion"
```

```
      ]
    }
  ]
}
```


D.2 Full per-utterance error analysis (150 utterances)

Total rows: 150. Categories are heuristic (jiwer alignment + regex), not mutually exclusive.

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
0685de35bc	8efbc	male	केवल तिब्बत	केवल तिब्बत	0.00	0.00	other
d33d533af8	8efbc	male	अहिलेको हुलाक भवनको	अहिलेको हुलाक भवनको	0.00	0.00	other
df8a0b212a	efa9c	male	हुने गर्थ्यो	हुने गरथ्यो	0.50	0.08	oov_rare_vocabulary
356225f520	07179	female	बाह्य मामलाको मन्त्रालय को	भाइदे मामलाको मन्त्रालय को	0.33	0.16	oov_rare_vocabulary, phonetic_confusion
dad71f7b34	1a81b	male	तयार पार्नु भन्दा	तयार पार्नुभन्दा	0.67	0.06	oov_rare_vocabulary
d2b4813104	12d45	female	बेलुका सुत्ने बेलामा	बेलुका सुत्ने बेलामा	0.00	0.00	other
7da0e9a7b7	99866	female	राज्यमाथि हासिल गर्‍यो	राज्यमाति हासेल गरयो	1.00	0.14	noise_degradation, oov_rare_vocabulary, phonetic_confusion
0f7a64d0bd	70e10	female	निबन्ध भित्रै पर्दछन्	निबन्ध भित्रै पर्दछन्	0.00	0.00	other

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
7edfb32bd1	c46b5	male	गर्ने प्रविधिमा हुने	गर्ने प्रविधिमा हुने	0.00	0.00	other
34f3dc6a6a	7f9c6	female	उनी गाउँको एउटा	उनी गाउँको एउटा	0.00	0.00	other
0ca2b67e85	e8f6f	male	यस सम्बन्धमा विशेष	यस सम्बन्धमा विशेष	0.00	0.00	other
3b99e9373e	da0cf	male	सेती हुन्	सेती हुन्	0.00	0.00	other
8e47838650	1a81b	male	महत्त्वपूर्ण स्थान	महत्त्वपूर्णै स्थान	0.50	0.06	oov_rare_vocabulary
8ccd499b56	da0cf	male	बनाउनबाट जोगाउँदछ	बनाउनबाटा जोगआउँदछ	1.00	0.12	noise_degradation, oov_rare_vocabulary, phonetic_confusion
89305ffa02	8efbc	male	अन्तिम युग हो	अन्तिम युग हो	0.00	0.00	other
8e908cd51d	07179	female	पिता पोषण पाण्डेको	पिटा पोषन पाण्डेको	0.67	0.11	oov_rare_vocabulary, phonetic_confusion
3856d8968f	8efbc	male	ड्रागनफल सँग मिल्दोजुल्दो सिउँडी	ड्रागनफल सँग मिल्दोजुल्दो सिउँडी	0.00	0.00	other
78cb6e474d	9d08c	male	पार्टीका	पार्टीका	0.00	0.00	other

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
0a524e5006	76dab	female	थिए एउटा ऐतिहासिक धार्मिक	थिए एउटा ऐतिहासिक धार्मिक	0.00	0.00	other
d399359741	efa9c	male	खोज बाकस बेकार छ	खोज बाकास व्यकार छ	0.50	0.25	oov_rare_vocabulary, phonetic_confusion
0cf3ebaf89	07179	female	उद्देश्यले ५ अप्रिल	उद्येश्यले पच्च अपरेल	1.00	0.26	noise_degradation, oov_rare_vocabulary, phonetic_confusion
3a523d2f4f	8efbc	male	गर्नका लागि छिमेकी	गर्नका लागि छिमेकी	0.00	0.00	other
0f5492d100	99866	female	ग्रस्त र अवसरवादी	ग्रस्त र अवसरवादी	0.00	0.00	other
09607fb527	8efbc	male	गुरुचेला धुमधाम झगडा	गुरुचेला दुमदाम चगडा	0.67	0.15	oov_rare_vocabulary, phonetic_confusion
0d3af9342e	056c7	female	समर्थनमा स्थापित गरिदिए	समर्थमा स्थापित गरिदिए	0.33	0.04	oov_rare_vocabulary
3cd1567940	e8f6f	male	ब्याज तिर्नुपर्ने हुन्छ	व्यहास दिन्नु पर्ने हुन्छ	1.00	0.26	noise_degradation, oov_rare_vocabul

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
							ary, phonetic_confusion
794dd07876	76dab	female	रोजा राखेर राति	रोजा राखेर राती	0.33	0.07	oov_rare_vocabulary, phonetic_confusion
3542203038	1a81b	male	जोडी नै सम्विधान सभामा पुग्ने सात जोडीमध्ये एक हो अमृताको जोडी पनि।	जोडिनैसं विधानस भामा पुग्ने साथ जोडीमध्य एक हो अमृताको जोडी पनि	0.55	0.12	oov_rare_vocabulary, phonetic_confusion
dc5bd13f97	e8f6f	male	नयाँ नेपाल	नयाँ नेपाल	0.00	0.00	other
dd07ccac9b	056c7	female	राजेश जी को	राजेश जीको	0.67	0.09	other
3b05885ecb	76dab	female	बाजा समूहको एक	बाजा समूहको एक	0.00	0.00	other
069f1a5975	9d08c	male	धेरै हुन्छ	धेरै हुन्छ	0.00	0.00	other
084f8770e0	07179	female	उमारिएको विभिन्न किसिमको	उमारिएको विभिन्न किसिमको	0.00	0.00	other
deff9009e8	da0cf	male	गुरिल्ला युद्धको लागि	गुरेल्ला युद्धको लागि	0.33	0.05	oov_rare_vocabulary, phonetic_confusion

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
0950a3e9a4	99866	female	रन्मामैकोट तकसेरा हुकाम	रणमा मैकोट तक्सेरा हुकाम	1.00	0.17	noise_degradation, oov_rare_vocabulary, phonetic_confusion
7b1f434978	99866	female	आइल्यान्ड तथा बेलायतदेखि	आइल्यान् तथा बेलाय देखि	1.00	0.08	noise_degradation, oov_rare_vocabulary, phonetic_confusion
06458f45c7	76dab	female	अञ्चलमा पर्ने काठमाडौँ	अञ्चलमा पर्नेकाठमाडौँ	0.67	0.05	oov_rare_vocabulary
d2d29b1aea	8efbc	male	रोकाको धारणा छ	रोकाको धारणा छ	0.00	0.00	other
068a5f9e4d	07179	female	साहित्य कृष्णप्रसाद पराजुलीद्वारा	साहित्य कृष्ण प्रसाध पराजुलीद्वारा	0.67	0.06	oov_rare_vocabulary, phonetic_confusion
3d23f3a053	056c7	female	चेतन तत्त्व छ	चेतन तत्यो छ	0.33	0.23	oov_rare_vocabulary, phonetic_confusion
8f20c18a27	7f9c6	female	नवप्रयोगहरूका कारण पनि	नवप्रयोगहरूका कारण पनि	0.00	0.00	other
8c693c9a97	7f9c6	female	मान्छे चटकारे	मान्छि चट कार्य	1.50	0.31	noise_degradation,

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							oov_rare_vocabulary, phonetic_confusion
8f61581d97	e8f6f	male	मानिससँग पनि	मानिसाँग पनि	0.50	0.08	oov_rare_vocabulary, phonetic_confusion
8e144a8c27	99866	female	चौतारी पृष्ठझैँ धेरै	चौतारी पृष्ठछैँ धेरै	0.33	0.05	oov_rare_vocabulary, phonetic_confusion
030127cb38	12d45	female	उनीहरूको कब्जा भयो	उनीहरूको कब्जा भयौ	0.33	0.06	phonetic_confusion
8a16923bc6	7f9c6	female	जातिका मनिसहरूको	जातिका मानिसहरूको	0.50	0.06	oov_rare_vocabulary
893dc361fd	8efbc	male	गुरुङको घरमा गयौँ	गुरुङको घरमा गयौँ	0.00	0.00	other
0a15894a63	70e10	female	आरम्भ भएको छ	अरम्व भएको छ	0.33	0.17	oov_rare_vocabulary, phonetic_confusion
0c0791cbc1	07179	female	सहयोग गर्न पाउनमा	सहियोग गर्न पाउनमा	0.33	0.06	oov_rare_vocabulary
d74c759f7d	efa9c	male	दिँदा पशु फाप	दिँदा पशु फाप	0.00	0.00	other
3a74cbb511	07179	female	मैले सुरुका दिनहरूमा	महैले सुरुका दिनहरूमा	0.33	0.05	oov_rare_vocabulary

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34760118f4	9d08c	male	वृक्ष यही हो	बृच्ष यहियो	1.00	0.42	noise_degradation, oov_rare_vocabulary, phonetic_confusion
dde677080a	70e10	female	सूर्यले पृथ्वीको वरिपरि	सवर्यलले फृथ्ष गरीपुरि	1.00	0.43	noise_degradation, oov_rare_vocabulary, phonetic_confusion
002a8d9076	76dab	female	र एक अन्य फिल्मको	र एक अन्य फिल्मको	0.00	0.00	other
3db3210f8e	5d5fe	male	उनलाई नचिन्नु	उनलाई नचि्नु	0.50	0.08	oov_rare_vocabulary
d8451ec160	99866	female	उत्तराधिकारी हुने	उत्तराधिकारी हुने	0.00	0.00	other
36e3b42ee0	76dab	female	दोलखा जिल्लाको	दोलखा जिल्लाको	0.00	0.00	other
81c40491fc	76dab	female	जिल्लाको पाठामारीमा अवस्थित	जिल्लाको पाठामारीमा अवस्थित	0.00	0.00	other
35ff798136	70e10	female	आग्नेय चट्टान संस्कृत शब्द	आग्ने चटटान संस्कृत शब्द	0.50	0.08	oov_rare_vocabulary
02443bae4d	1a81b	male	उद्योग कार्यालय	उद्योग कार्यलाय	0.33	0.11	oov_rare_vocabulary

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8ebb1af0c6	5d5fe	male	हो यो तथ्य इतिहासहरूमा	हो यो तथ्य हितिहासहरूमा	0.33	0.11	oov_rare_vocabulary, phonetic_confusion
8dae7b59b8	056c7	female	भए जहाँ भोजनको	भएजहाँ भोजनको	0.67	0.07	oov_rare_vocabulary
dfea2d149d	99866	female	गराएको मानिन्छ	गराएको मानिन्छ	0.00	0.00	other
d34cb037a8	e8f6f	male	प्रमुख थियो	प्रमुख थियो	0.00	0.00	other
d1cfb4aabc	efa9c	male	रचना समेत गरे	रचना समेट गरे	0.33	0.08	oov_rare_vocabulary, phonetic_confusion
0e8987f3fc	5d5fe	male	एतिहासिक प्रशासनिक महत्त्वको	ऐतिहासिक परासानिक महत्त्वको	0.67	0.14	oov_rare_vocabulary, phonetic_confusion
045f34cd5a	c46b5	male	तपाईँ आफ्नो गाउँको	तपाईँ आफ्नो गाउँको	0.00	0.00	other
3b88864e56	5d5fe	male	राष्ट्रिय सेवाहरू	राष्ट्रिय सेवाहरू	0.00	0.00	other
361164a7ed	70e10	female	स्थलमध्ये पर्दछ	स्थनबध्ये पर्दछ	0.50	0.13	oov_rare_vocabulary, phonetic_confusion
da5d1a8a89	e8f6f	male	धनकुटाको कचिडेमा	धनकुटाको कचिडेमा	0.00	0.00	other

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84efea6b34	76dab	female	अमेरिकी डलर छ	अमेरिकी डलर छ	0.00	0.00	other
dfc8d36ad2	efa9c	male	जहाँ दुखहरूको अग्नि	जहाँ दुखहरूको अग्नी	0.33	0.05	oov_rare_vocabulary, phonetic_confusion
8aa1a58431	12d45	female	जरूरी छैन जुन	जररी छैन जुन	0.33	0.08	oov_rare_vocabulary
061f10be2e	70e10	female	आफ्नो क्षमता	आफ्नो क्षमता	0.00	0.00	other
8c3844dbb0	efa9c	male	स्टारकिड को पहिलो राष्ट्रिय	स्टारकिड को पहिलो राष्ट्रिय	0.00	0.00	other
7921dc943c	efa9c	male	नामक गणितसम्बन्धी पुस्तक	नामक गणित सम्बन्धी पुस्तक	0.67	0.04	oov_rare_vocabulary
d9aa2f3129	efa9c	male	कारण एकै समयमा	कारण एकै समयमा	0.00	0.00	other
3f618a3c25	1a81b	male	जिल्ला पञ्चायतहरूमा निर्वाचित	जिल्ला पञचतहरूमा निर्वाचित	0.33	0.10	oov_rare_vocabulary
3161b117c2	1a81b	male	आउने बित्तिकै क्रिस्टल	आउने भित्तिकै कृष्टल	0.67	0.23	oov_rare_vocabulary, phonetic_confusion
7a01a55c6a	8efbc	male	चरनयोग्य क्षेत्र मानिन्छन्	चरण योग्ञ क्षेत्र मानिन्छन्	0.67	0.12	oov_rare_vocabulary, phonetic_

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
							confusion
356db48125	e8f6f	male	नोकियाको मोबाइल छ	नोकियाको मोबाइल छ	0.00	0.00	other
d61b84ec00	7f9c6	female	माध्यमिक विद्यालयको	माध्यमिक विद्यालयको	0.00	0.00	other
d0940523bb	8efbc	male	राम्रो व्यक्ति बन्नेमा	राम्रो व्यक्ति बन्नेमा	0.00	0.00	other
7752c1a8fa	99866	female	होल्टका भावनाहरू र सन्	होल्डका भावनाहरू र सन्	0.25	0.05	oov_rare_vocabulary, phonetic_confusion
3295ade169	76dab	female	परयाहरूको संजालभित्र	परयाहरूको सञ्जालभित्र	0.50	0.15	oov_rare_vocabulary, phonetic_confusion
8921260225	f07db	female	मनुको हो	मनुको हो	0.00	0.00	other
d443c84109	70e10	female	क्रान्तिकारी वाममोर्चाको	तन्थिकारी बाम वर्षको	1.50	0.42	noise_degradation, oov_rare_vocabulary, phonetic_confusion
3f8bcfc6b9	99866	female	एक अनुमान छ	एक अनुमान छ	0.00	0.00	other
d72cfed93c	c46b5	male	साउथ अस्ट्रेलियाली	साउथ अस्ट्रेलयाली	0.33	0.04	oov_rare_vocabulary

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73fde60c68	e8f6f	male	चलचित्र भिक्षा दिनका निम्ति	चलचित्र भृक्षा दिनका निम्ति	0.33	0.05	oov_rare_vocabulary, phonetic_confusion
d20b2fea89	056c7	female	निसी नेपालको	निशी नेपालको	0.50	0.08	oov_rare_vocabulary, phonetic_confusion
7f3e582176	8efbc	male	तिनै हजुरबुबाको आडमा	तिनै हजुर बुभाको आढमा	1.00	0.15	noise_degradation, oov_rare_vocabulary, phonetic_confusion
3270c486a1	f07db	female	गरेको तपाईँको योगदान	गरेको तपाईँको योगदान	0.00	0.00	other
0e8819f2e1	da0cf	male	बाँकी नै छ	बाकी नै छ	0.33	0.10	oov_rare_vocabulary
dc1c739b7d	f07db	female	धर्म सम्प्रदायहरू जस्तै	धर्म समप्रदायहरू जस्तै	0.33	0.04	oov_rare_vocabulary
88f3036a36	99866	female	अङ्ग्रेजी महिना हो	अङ्ग्रेजी महिना हो	0.00	0.00	other
80d51d2e79	da0cf	male	बेलायती प्रभावमा कमी	बेलायती प्रभावमा कमी	0.00	0.00	other
79ecb6ff25	1a81b	male	किम कुलिग	किम कुलिकक	0.50	0.22	oov_rare_vocabulary,

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
							phonetic_confusion
d54cb86f18	f07db	female	बिनायो तथा हारी	बि्ना ययो तथा हाली	1.00	0.27	noise_degradation, oov_rare_vocabulary, phonetic_confusion
0aead87b58	70e10	female	ढल्कनासाथ देवताहरूको	ढल्कन सत्देवतहरूको	1.00	0.25	noise_degradation, oov_rare_vocabulary, phonetic_confusion
0d23be1ca8	70e10	female	यस अवसरमा	यस औसरमा	0.50	0.22	oov_rare_vocabulary, phonetic_confusion
df0f134997	c46b5	male	९ गते सिरहा	[UNK]ौ गतेशीरहा	1.00	0.82	noise_degradation, oov_rare_vocabulary, phonetic_confusion
01af564842	8efbc	male	प्वाँख मैले	पवाँख मैले	0.50	0.09	oov_rare_vocabulary
8f268b2d42	f07db	female	हुन स्वीकार	हुन स्विकार	0.50	0.09	oov_rare_vocabulary, phonetic_confusion
0eace1a98b	056c7	female	पढाइ प्रतिको	पढाइपर्ति को उहाँको	0.67	0.16	oov_rare_vocabulary,

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			उहाँको				phonetic_confusion
7635799596	12d45	female	मात्र रिडाइरेक्ट हुन्छ	मात्र रिडाइरेक्ट हुन्छ	0.00	0.00	other
3dcd687720	efa9c	male	बिमलादेवीको सानो चिटिक्क	बिमलादेवीको सानु चिटिक्क	0.33	0.04	oov_rare_vocabulary, phonetic_confusion
81f4c30a2c	99866	female	कि खतरा	कि खतरा	0.00	0.00	other
725c6c0064	e8f6f	male	काम गर्दागर्दै वहाँ	काम गर्दा गर्दै उह	1.00	0.21	noise_degradation, oov_rare_vocabulary, phonetic_confusion
dd26100ea3	12d45	female	नाम जे राख्नुहोस्	नामजे राख्नु होस्	1.00	0.12	noise_degradation, oov_rare_vocabulary
0d95694cfe	efa9c	male	उपल्लो अर्खला नवलपरासी	उपङ्लु अर्खला नबल परास	1.00	0.27	noise_degradation, oov_rare_vocabulary, phonetic_confusion
0230c928e0	f07db	female	गङ्गालाल श्रेष्ठलाई गोली	गङ्गलाल श्ेष्ठलाई बोली	1.00	0.12	noise_degradation, oov_rare_vocabulary,

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
							phonetic_confusion
770279b652	99866	female	मित्रता निकै	मित्रता निकै	0.00	0.00	other
8a8b3a56d5	7f9c6	female	ठूलो यज्ञ गरे	ठुलो एज्य गरे	0.67	0.23	oov_rare_vocabulary, phonetic_confusion
7d7bdec74f	5d5fe	male	नेपालमा राणाहरूले	नेपालमा राँडाहरूले	0.50	0.12	oov_rare_vocabulary, phonetic_confusion
86755b50ef	5d5fe	male	सिरलिङ्गे लिएर विभिन्न	सिरलिङ्गे लिएर विभिन्न	0.00	0.00	other
d4b24720b7	76dab	female	पराग एउटा नेपाली	पराग एउटा नेपाली	0.00	0.00	other
378bb42091	8efbc	male	मातृभाषा वा निजहरूको	मात् भाषा वा निजहरूको	0.67	0.10	oov_rare_vocabulary, phonetic_confusion
08606b64cc	8efbc	male	पिर मर्का गुनासाहरू	पीरमरका गुनासाहरू	0.67	0.16	oov_rare_vocabulary, phonetic_confusion
7b9949c9e6	76dab	female	विजय	विजय	0.00	0.00	other
382856c61f	12d45	female	मेरो मुखबाट एउटै	मेरो मुखबाटै एउटै	0.33	0.06	oov_rare_vocabulary
84b58e62	70e10	female	स्थानीय	स्थानी	0.67	0.19	oov_rare

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23			उद्योगका लागि	उद्वोकय लागि			_vocabulary, phonetic_confusion
0c2a6969f8	07179	female	यसलाई अङ्ग्रेजीमा रिभर	यसलाई अङ्ग्रेजीमा रिबर	0.33	0.05	oov_rare_vocabulary, phonetic_confusion
0139dc5b13	056c7	female	बनाएर आधुनिक गीत	बनाएर आधुनिक गीत	0.00	0.00	other
8a24eda5e8	da0cf	male	पार्टीको सहयोगको लागि	पाटीको सहयोगको लागि	0.33	0.10	oov_rare_vocabulary
0176bd214c	9d08c	male	मुगल शासकुन्ले	मुगल शासककुनले	0.50	0.14	oov_rare_vocabulary
3a26f35fc1	76dab	female	जापानमा बनाइएको एक	जापानमा बनाइएको एक	0.00	0.00	other
05ec166b72	5d5fe	male	नेपालको उपाध्यक्ष साथै	नेपालको उपादक्ष साथै	0.33	0.14	oov_rare_vocabulary, phonetic_confusion
7d68ea40c7	70e10	female	हजार ८०० मिटरदेखि	हज[UNK] [UNK] [UNK]मिटरदेखि	1.00	0.88	noise_degradation, oov_rare_vocabulary, phonetic_confusion
8ab6e41132	5d5fe	male	कसैले कुनै	कसैले कुनै	0.00	0.00	other

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78c9e55641	efa9c	male	पनि जाने इच्छालाई समाप्त	पनि जाने इच्छालाई समात	0.33	0.10	oov_rare_vocabulary
7b86a1f14e	1a81b	male	मई १९७४ मा	मै [UNK] [UNK] [UNK] [UNK]मा	1.00	2.10	noise_degradation, oov_rare_vocabulary, phonetic_confusion
ddb44bb107	8efbc	male	भक्तपुरपछिको ठूलो प्राचीन	भक्तपुरपछिको ठुलो प्राचीन	0.33	0.04	oov_rare_vocabulary, phonetic_confusion
3cb1356eec	8efbc	male	हो जहाँबाट हामीले	हो जहाँबाट हामीले	0.00	0.00	other
dfaf3cfaf5	e8f6f	male	परिभाषाले स्पष्ट पारेको छ	परिवाषाले स्पस्त परादेको छ	0.75	0.20	oov_rare_vocabulary, phonetic_confusion
d0480f7916	056c7	female	ठोस ज्यामितिका सन्दर्भमा	ठोस ज्यामितिकका सन्दर्वमा	0.67	0.08	oov_rare_vocabulary, phonetic_confusion
0584cc526e	da0cf	male	छतौल नेपालको जनकपुर	छतौल नेपालका जनकपुर	0.33	0.05	oov_rare_vocabulary, phonetic_confusion
8b50556041	f07db	female	उपन्यास कृष्णहरि	उपन्यास कृष्णहरि	0.00	0.00	other

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
0e8a19bda9	8efbc	male	बराल भेडा परेवा हाँस	बराल बेडा परेवा हास	0.67	0.13	oov_rare_vocabulary, phonetic_confusion
75acd88ff6	056c7	female	परेको र आफू बालकै	परेको र आफूवालककै	0.50	0.18	oov_rare_vocabulary, phonetic_confusion
0a5ad7b85e	c46b5	male	सबैले एकमतले स्वीकारेको	सबैले एकमतले स्विकार्यको	0.33	0.13	oov_rare_vocabulary, phonetic_confusion
77db4a7462	5d5fe	male	तर पनि त्यस	तर पनि त्यस	0.00	0.00	other
d8814e3d67	9d08c	male	पवित्र मानिन्छ	पवित् र मानिन्छ	1.00	0.07	noise_degradation, oov_rare_vocabulary
06330c8dde	99866	female	यस भवनको निर्माणले	यस भवनको निर्माणले	0.00	0.00	other
8f24eac8cf	1a81b	male	तपाईँले बिना सन्दर्भका	तपाईँले विनास अन्दर्वका	0.67	0.18	oov_rare_vocabulary, phonetic_confusion
d7d0184f51	efa9c	male	विष तयार गर्छ	बस्तयार वर्ष	1.00	0.46	noise_degradation, oov_rare_vocabulary, phonetic_

Utterance ID	Speaker	Gender (pseudo)	Reference	Hypothesis	WER	CER	Categories
							confusion
7553b80c6b	5d5fe	male	शाहको नामबाट तत्कालीन	शाहको नामबाट तत्कालीन	0.00	0.00	other
09608a650b	056c7	female	मुद्रा जनसङ्ख्या क्षेत्रफल	मुद्रजनसङ्ख्या क्षेत्रफल	0.67	0.08	oov_rare_vocabulary
3cf2fad57c	8efbc	male	पुरस्कार पाउनुभएको थियो	पुरस्कार पाउनुभएको थियो	0.00	0.00	other
8d4a047584	efa9c	male	विद्यालय बनेको हो	विद्यालय बौनेको हो	0.33	0.06	oov_rare_vocabulary